

# Powering Power Electronic Simulations using PSIM

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In this article we will discuss about PSIM, a fast, yet accurate, simulation environment for power electronic designs, motor drive simulations and energy-conversion applications. The software is positioned midway between a SPICE-like device or circuit simulator and a system or control simulator like MATLAB or Simulink. PSIM enables the user to build circuits and obtain results easily even without any previous experience of using computer aided design (CAD) software.

The software is widely used in industry for research and development of new products, while academicians use it for research and teaching. The demo of the latest release of this software, version 10.0.2, is included in this month's DVD for EFY Plus.

## Why PSIM

This software allows designers to simulate power circuits with any level of detail. The initial version of PSIM was originally released with ideal components only. In the newer version there are only a limited number of equations to be solved, and hence the speed of simulation could be fast.

**Simulating power circuits of varying complexities.** With the latest version of PSIM it is now very easy to selectively introduce parasitic interactions and non-idealities. This is the key as it allows you to introduce more complex problems as you are ready to solve these, as opposed to

how some simulation packages force you to deal with complex issues before you have even settled on major specifications.

PSIM is considered among the fastest simulation environments. It uses a strong algorithm dedicated to electrical circuits (piecewise method, generic models and fixed time-step). The fast simulation allows repetitive simulation runs and significantly shortens the design cycle.

There is no learning curve; one can build circuits in PSIM and start

## Quick view

**Latest version:** 10.0.2, March 2015

**Developers:** Powersim

**Operating system and processor architectures:**

- Microsoft Windows 7 and 8
- All latest versions run on 64-bit machines
- Version 9 is available as both 32-bit and 64-bit

**Type:** Simulation software

**Website:** [www.powersimtech.com](http://www.powersimtech.com)

seeing waveforms in minutes. In addition, with a robust solver, PSIM has no convergence issues that other simulation software face.

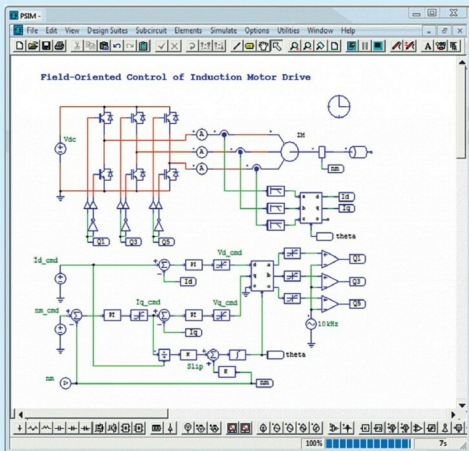


Fig. 1: PSIM simulation environment